

Transforming health care through Bhutan's digital health strategy: progress to date

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Abstract

Bhutan, a landlocked country in the eastern Himalayas with some of the most rugged and mountainous terrain in the world, is actively engaged in digital health strategy reforms aimed at improving the efficiency of the health information system. Aligned with Bhutan's *e-Government master plan*, the *National eHealth strategy and action plan* aims to improve health by empowering health-care providers and citizens through technology and by enabling data exchange for service delivery. The strategy has four primary areas of focus: (i) ensuring digital health governance arrangements; (ii) concentrating on strong foundations in terms of infrastructure and standards; (iii) prioritizing improvements in the current health system in a phased, selective manner; and (iv) building the digital skills and knowledge of health workers. With support from the Asian Development Bank and the World Health Organization, phase 1 of the strategy has been completed and the blueprint for the digital health information system is in development. Phase 2 of the strategy will be implemented during 2020–2023 and will include work on (i) identity management for the health workforce; (ii) the implementation of a master patient index and a secure longitudinal patient information system; and (iii) enabling all health facilities to access the systems. Bhutan's eHealth strategy has the potential to fundamentally transform the delivery of health services, strengthen primary health care and enable the development of a "One Health" public health surveillance system.

Keywords: Bhutan, digital health, electronic health records, health information system, health information technology

Background

Bhutan is a landlocked South-East Asian country in the eastern Himalayas with some of the most rugged and mountainous terrain in the world and a population of around 736 000, according to the census in 2017.¹ The health indicators in Bhutan are among the best in the low- and middle-income countries; in 2017, average life expectancy was 70.2 years, sustained child immunization coverage was above 95% and household out-of-pocket expenditure on health was low.^{1–3} The Royal Government of Bhutan provides a comprehensive range of free health services to all its citizens. In 2015, Bhutan spent 3.6% of its gross domestic product on total health expenditure, including expenditure on the largely publicly financed and managed health-care system,² which includes free air travel for citizens seeking health care not available in the country.² Despite the relative generosity of the government, there are concerns about the sustainability of health finances, and there has been a realization that methods of reducing waste and increasing efficiency are needed. This is especially urgent because the country is facing a growing double burden of communicable and noncommunicable diseases, including

challenges such as increases in substance use, suicide and other mental health problems.⁴

A key component in increasing the resilience of a health system to new challenges is improving and upgrading the health information system (HIS).⁵ Management and governance of health information in Bhutan are, overall, highly disjointed.⁶ Fragmentation in HISs within the Ministry of Health is a particular challenge, preventing efficient data sharing and analysis. This has direct and indirect effects on all aspects of health care. For example, the lack of an accurate, timely and reliable patient information system makes it difficult to track and refer patients and to coordinate the continuum of care between primary health care and higher-level hospital care.⁷ A particular problem is reliance on paper-based medical records, which is resource intensive and often inefficient in monitoring, evaluating and delivering optimal health services. For instance, the inordinate complexity of quantifying the burden of patient care when relying on paper-based records can be a barrier to the efficient provision of health services, which a recent study has shown to be suboptimal.⁸

The Royal Government of Bhutan has recognized the potential of information and communication technology (ICT)

to have a positive impact on health care and in 2018 launched the *National eHealth strategy and action plan*.⁹ This perspective paper provides a report on progress to date on digital health strategy reforms aimed at improving the efficiency of the HIS in Bhutan.

National eHealth strategy and action plan

The mission of the *National eHealth strategy and action plan* is to support the provision of better health care by empowering health-care providers and citizens through technology and by enabling data exchange.⁹ The strategy was developed with support from the Asian Development Bank and the World Health Organization,¹⁰ based on a national digital health vision, action plan and monitoring framework. The chronology of the reforms that led to the current aspirations with regard to digital health reforms in Bhutan is shown in Table 1.

The digital health strategy reforms aim to improve the efficiency of the HIS in Bhutan by refining governance mechanisms, supporting patient management and care delivery at all health facilities, and improving the efficiency of disease surveillance through the “One Health” approach. The goal is to deliver sustainable, free and high-quality health-care services closer to the doorstep of every citizen. The 2018 *National eHealth strategy and action plan* has four primary areas of focus: (i) ensuring digital health governance arrangements; (ii) concentrating on strong foundations in terms of infrastructure and standards; (iii) prioritizing improvements in the current health system in a phased, selective manner; and (iv) building the digital skills and knowledge of health workers.⁹ The overall budget for the current phase is US\$ 52.5 million.⁹

Phase 1

Phase 1 of the digital health strategy was implemented during 2018–2019. This phase comprised the following components.

Governance

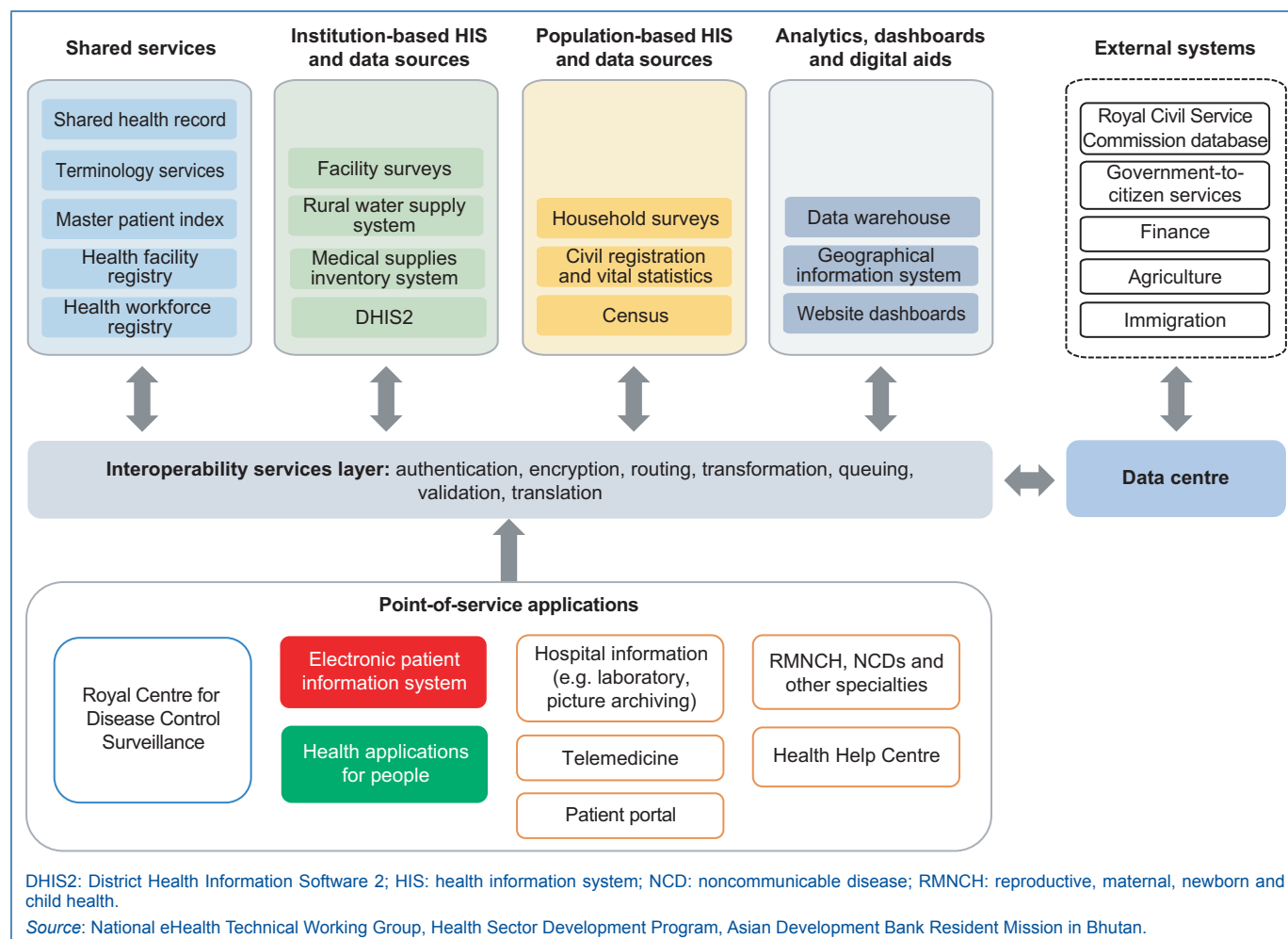
A steering committee was established, comprising 12 permanent members under the chairmanship of Dasho Secretary, Ministry of Health. The committee's role is to evaluate, direct and monitor the implementation of the strategy and action plan to ensure that the expected benefits are realized, with a reduction in risks and resources optimized. The committee also approves new digital health projects to ensure interoperability and sustainability. Critical leadership is provided by a technical working group with overall responsibility for creating an ecosystem addressing people, data and system issues, in alignment with the strategic objectives for the health sector.

Infrastructure development and integration

The eHealth enterprise architecture – the blueprint for the HIS – is currently in development and will encompass IT infrastructure development, data integration and technical standards for health data exchange.¹⁵ Both the *National health policy*¹⁶ and the *Healthcare ICT master plan*¹³ advocate an electronic medical records system in every health facility. The first iteration of this information system, along with an electronic patient information system, has been completed by Bhutanese experts. Strong technical leadership is required in developing the architecture for a state-of-the-art strategic information system. It is envisaged that the electronic health records will retrieve data prospectively, ensuring patient monitoring across several providers from different information systems integrated with the District Health Information Software 2 (DHIS2). The core platform, a centralized national health data warehouse, will link data from multiple sources and present it in a standard format via the electronic portal, thereby ensuring that all aspects of health data are interoperable. These sources will both feed data into and be able to extract data from the national warehouse (see Fig. 1).

Table 1. Timeline of digital health reforms in Bhutan, 1984–2023

Year(s)	Initiative	Outcomes/goals
1984–2002	Bhutanese Health Information Unit ³	Data on morbidity and mortality from health facilities collected and compiled by hand
2003–2013	Transition from a paper-based to an electronic database of morbidity and mortality data ³	Stand-alone database compiled in Microsoft Access
2006	Bhutan Information, Communications and Media Act 2006 ¹¹	Transition from Microsoft Access to the current web-based system initiated
2014	<i>e-Government master plan</i> ¹²	Ensuring universal access to safe and secure government e-services by facilitating the provision of citizen-centric services
2014	<i>Healthcare ICT master plan</i> ¹³	Strengthening fragmented systems in terms of infrastructure, internet connectivity, human resources, financial investments, data collection and reporting, and integration
2014–2017	Implementation of District Health Information Software 2 (DHIS2) ¹⁴	Web-based aggregate health data collection established using DHIS2, a free open-source health management data platform
2018–2019	Phase 1 of the national eHealth strategy	- Select and prepare systems to be improved in years 3–5 - Deliver low-risk, low-cost eHealth services - Prepare for the long term with a focus on identifiers, standards and networks - Capacity-building
2020–2023	Phase 2 of the national eHealth strategy	- Maintain and increase momentum from phase 1 - Roll out key identifiers and standards and upgrade networks - Implement key improvements in the more complex systems - Continue capacity-building

Fig. 1. Planned digital health architecture for Bhutan

Services and applications

Bhutan aims to deliver low-risk, low-cost digital health services to its citizens to achieve universal health coverage. Health-care providers with smartphones will be assisted by specially designed mobile apps and knowledge support systems under a central licensing arrangement. Improvements in routine web-based data collection and the implementation of interoperable technologies have already enhanced both disease surveillance and health security in the country. For example, Bhutan's National Early Warning Alert and Response Surveillance system enables reporting to the Royal Centre for Disease Control through an easy-to-use web-based information system.¹⁷

Workforce capacity development

The Ministry of Health has assessed the ICT education needs. Training programmes for all health professionals to increase capacity and fill specific digital health gaps are planned. The training will be delivered through online modules and apps, and audiovisual materials, and will include continuing medical education. In addition, online educational and behaviour-change communications will be used to target patients and citizens.

Phase 2

Phase 2 of the digital health strategy will be implemented during 2020–2023, with a focus on maintaining and increasing the momentum of digital health reforms. Further work includes identity management for the health workforce, the effective implementation of a master patient index, facilitation of longitudinal health records, clinical system linkage and improved access to patient health information. By 2023, it is expected that web services offering interoperability among platforms will be extended to other services such as the Health Help Centre for health information and ambulance services, the national blood transfusion service, and facilities such as district hospitals, primary health-care units and outreach clinics. Training of the current and new cadres of digital health experts will be sustained through online training programmes.

Future directions in digital health for Bhutan

The digital health strategy is ambitious and requires sustained governance and management commitment, user support and

the collective effort of all stakeholders. These stakeholders include government entities such as the Cabinet, the Gross National Happiness Commission, the Royal Civil Service Commission, the National Institute of Traditional Medicine Services and the administrations of the dzongkhags (provinces). Stakeholders at the Ministry of Health level include the Jigme Dorji Wangchuck National Referral Hospital, the hospital network and the basic health units. The primary goal of the digital health strategy is to enable increased access to quality health care for all. The plan is to achieve this by influencing positive changes in health-related knowledge, behaviour and practice among the public, health-care professionals and policy-makers. The use of an information management system should improve health-related decision-making and thus perceptibly improve the quality of integrated and timely care of patients.

The Royal Government of Bhutan aims to integrate all the digital services related to health-care service delivery into a single efficient platform. As previous case-studies have shown, robust planning will be essential to ensure successful implementation and that consideration is given to factors influencing adoption, such as cultural norms, management structures and the adaptability of users.¹⁸ For example, research on cybersecurity in Bhutan has highlighted the need for government organizations to create a culture of security among all IT users as a means of protecting the confidentiality, integrity and availability of information systems and networks.¹⁹ A systematic review of evaluations of health data management found that poor quality of health data, low levels of health information use and poor management were the main hindrances to the implementation of HISs in low- and middle-income countries.²⁰ The effectiveness of lay health workers depends on how quickly and efficiently they can adapt to HISs, and, therefore, is critical to any attempt to integrate data collection platforms into HISs.²¹ In addition, the sustainability of finances for maintaining and improving HISs remains a key challenge in most low- and middle-income countries.

There is mixed evidence of the success of HIS interventions in low- and middle-income countries. Electronic health records as the core clinical application in an HIS can prove to be challenging, owing to the intricacies involved in recording data and a lack of scientific rigour.^{22,23} However, changing from pen and paper to the use of personal digital assistants to collect data in a survey across 21 000 scattered rural households in the south of the United Republic of Tanzania saved time and improved the quality of the data recorded.²⁴ There is clear evidence that the use of personal digital assistants, tablets or mobile devices improves the timeliness of data collection and the quality of the data.²⁴ Similarly, separate studies on antenatal care in Indonesia have shown a willingness on the part of midwives to adopt digital methods²⁵ and a positive response to the use of electronic pregnancy registers that improved capture of routine antenatal data.²⁶ Using open-source platforms to collect data on more than 25 000 patients for a complex randomized controlled trial in low-income settings in Nepal proved efficient and feasible.²⁷ A review of Population Health Implementation and Training Partnerships in five sub-Saharan African countries suggests that the best HISs not only are flexible but also follow an iterative approach in designing and refining the development of new tools and

approaches.²⁸ Therefore, rigorous trials, routine data-quality audits and troubleshooting, incorporating timely feedback to improve HISs based on health system performance, are key attributes in the design and development of HISs. Investments in data audits and feedback activities intended to improve HIS data have resulted in significant improvements in data quality in low- and middle-income countries such as Mozambique.²⁹ Health systems in low- and middle-income countries need internal mechanisms to develop performance targets, track progress, and create and manage knowledge for continuous improvement. Using evidence-based recommendations such as the Performance of Routine Information System Management (PRISM) framework can help considerably in designing, strengthening and evaluating an HIS.³⁰ Frameworks such as this can help in focusing on performance, incorporating organizational and behavioural determinants, and ensuring accountability in HISs.³⁰

The implementation of the eHealth strategy and interoperable HIS is reflected in the forthcoming Royal Government of Bhutan Gross National Happiness Commission 12th 5-year plan (2018–2023).³¹ The focus of the HIS is developing ICT-literate cadres in the health workforce by improving workers' learning and competencies, and encouraging behaviour change. Confronted with a progressive decline in overall external assistance to the country, the Royal Government of Bhutan is striving to generate resources to sustain the digital health initiatives and incremental reforms. The Asian Development Bank, as a part of its wider health sector support, is also providing budgetary support to the government to drive its agenda with regard to improving strategy and governance; information technology infrastructure, services and applications; data standards for interoperability; and workforce capacity development. Given the substantial investments being made in digital health, it is essential to quantify the impact and outcomes. A monitoring and evaluation framework should be put in place to assess progress and identify where improvements in the delivery of projects and systems can be made.

Concerns have been reported regarding the use of electronic health records (i) prolonging working hours because of the time spent on tasks such as data entry;^{32,33} and (ii) reducing professional satisfaction because of issues such as time directed away from patient interaction and the inflexibility of using templates for notes.^{34–36} It is, therefore, essential that the desired eHealth outcomes are defined and measured to inform improvements to the system. In addition, it is vital that there is a focus on the citizen-centricity of the health-care system, to enable the end-users to actively take control of their health and well-being. In the implementation phase, it is important to deal with barriers to the adoption of digital health services, by ensuring that the finances are sustainable^{37,38} and by addressing users' concerns.^{38,39}

Through mobile apps and tracking systems, personal health information management can be implemented in creating and sustaining healthy communities. Furthermore, digital health platforms should focus on catching up with the dynamic pace of technological advances and the proliferation of online social networks. The full impact of digital health can be realized beyond its direct clinical impact; for example, it can facilitate preventive, long-term outpatient care. Developing digital health solutions for complex health systems is an ongoing

cyclical process, guided by health system strategies and policies. Multisectoral collaboration is essential for the success of a programme involving several stakeholders, including laypersons, health-care providers, health service managers, policy-makers, researchers and donors.

Conclusion

Bhutan envisages the nationwide deployment of an advanced HIS that will significantly scale up the implementation of the Ministry of Health's ambitious digital health strategy. The current partners are committed to supporting infrastructure solutions, migration planning and the implementation of governance mechanisms to bring about these reforms. If successfully implemented, Bhutan's HIS has the potential to improve patient safety outcomes, inform policy and transform health-care services. Thus, the Royal Government of Bhutan is on the path towards implementing a streamlined, interoperable, comprehensive HIS.

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